

Alex Lee

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EDUCATION

University of New South Wales

Sydney, Australia

Bachelor of Computer Science

May 2022 – Feb 2027

- WAM: 81 (2025 WAM: 87) – approx. 3.8/4.0 GPA (2025 GPA: 4.0/4.0)
- Received C++ Graduate Software Engineer offers for Susquehanna (SIG) and Vivcourt Trading
- Computer Science Mentor for Colombo House (2024), Trainee Lead for UNSW Development Society (2025, 2026)
- UniHack 2025 (Largest ANZ Hackathon): Best Design Award & Social Impact Award: [Link](#)

EXPERIENCE

Software Engineer Intern

Nov 2025 – Feb 2026

Google

Sydney, Australia

- Google Maps, Geo-Data Team
- Independently researched novel signals across petabytes of raw Google Maps edit history and user reports that correlate with incorrect edits, engineering pipelines to detect regressions and ensure data integrity.
- Developed validation frameworks in Java, SQL and Go to compare pre-edit and post-edit signals, flagging 0.4% of all Google Maps edits as anomalies and automating rollbacks for high-risk corrupted production data.
- Received a Return Offer after thorough reviews of positive team feedback, code quality and documentation.

Software Engineer Intern

Jun 2025 – Nov 2025

TikTok

Sydney, Australia

- Trust & Safety, Engagement Safety (Direct Messages)
- Enhanced Python QA frameworks to test Go microservices and drove DM Safety stability & coverage metrics from 6th to 1st out of 14 Trust and Safety subteams, maintaining RPC test stability rates of 99%+ every week.
- Scaled testing infrastructure, developed internal tools, created testing pipelines for the stability of millions of TikTok DM users, cutting cron-job test runtimes by 81%.

Tutor (Teaching Assistant)

Feb 2024 – Present

UNSW Computer Science & Engineering

Sydney, Australia

- Taught 300+ students and re-wrote legacy labs & assignments as a senior tutor for COMP2041.
- COMP2041 (Software Construction [*Linux, Shell, Python*]) Tutor: Overall rating 5.88/6.00 (avg: 5.39/6.00).
- COMP3311 (Database Systems [*SQL, Database Design*]) Tutor: Overall rating 5.67/6.00 (avg: 5.39/6.00).

PROJECTS

q-lite | *C/C++, Operating Systems, High-Performance Computing*

- Made a columnar time-series database in C++20 inspired by kdb+, using a Structure-of-Arrays layout with tagged union type dispatch to maximise L1/L2 cache utilisation during column scans while avoiding vtable overhead.
- Implemented a custom mmap-backed arena allocator with 64-byte aligned data placement for AVX2 compatibility, achieving 7x higher allocation throughput compared to standard `malloc` and `new`.
- Wrote SIMD-accelerated column operations using AVX2 intrinsics with streaming stores for large datasets, and a cache friendly open-addressing hashmap with linear probing achieving 3x throughput over `std::unordered_set`.

SmartTransit | *Python, SQL, AWS, Algorithms, Data Structures, Flask, Stripe, Docker, React Native*

- Created a bus transportation app where bus routes dynamically change according to real-time customer needs.
- Implemented backend infrastructure on AWS using Dockerised Flask services + RDS, with CI/CD deployment pipeline and basic monitoring/logging to support production releases, with Stripe and Google Maps API.

Redis Clone | *C/C++, Socket Programming, Networks, Data Structures, Operating Systems, Redis*

- Built a high-performance Redis clone from scratch, featuring a non-blocking, event-driven architecture using polling for scalable multi-client handling, efficient I/O, and minimal syscall overhead
- Implemented custom protocols for client-server communication, with RESP-style command parsing.
- Explored various client connection models, including multithreading, event-driven polling, and WebSockets to compare scalability, latency, and system resource usage.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Shell, Go, SQL, JS/TS

Systems: Operating Systems, Database Systems, Compilers, Computer Networks, HPC, Distributed Systems

Misc: Data Structures, Algorithms, Object-Oriented Programming, System Design, Git, gRPC, Protobuf, CI/CD, AWS